

NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

PYTHON FOR DATA ANALYTICS - PROJECT REPORT

Title: Student Performance Analytics

1. ABSTRACT

This project analyzes a student performance dataset using Python in Google Colab. The objective is to identify factors affecting student academic performance and generate useful insights. The dataset was cleaned, processed, and analyzed using Pandas, NumPy, Matplotlib, and Seaborn. Exploratory Data Analysis (EDA) techniques were applied to understand score distributions, correlations, and trends among different student groups. Visualizations were created to support decision-making. The analysis revealed patterns that can help educators improve student outcomes.

2. PROBLEM STATEMENT

Educational institutions need to understand the factors influencing student performance. Without proper analysis, it is difficult to identify learning gaps and improve academic results. Data analytics helps uncover trends and supports better educational planning.

3. OBJECTIVES

Generate meaningful insights from student data, identify performance patterns, and support data-driven decisions.

4. PROJECT SCOPE

Analyze student performance data using Python and visualize findings through charts and graphs.

5. TECHNOLOGIES USED

Development Environment: Google Colab
Programming Language: Python
Libraries: Pandas, NumPy, Matplotlib, Seaborn
Dataset Type: CSV

6. APPLICATION FLOW

Load dataset → Data cleaning → Data preprocessing → Exploratory Data Analysis → Visualization → Final insights.

7. KEY PYTHON DATA ANALYTICS CONCEPTS USED

Data Handling, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization.

8. EXPECTED OUTPUT

Dataset preview, cleaned dataset, performance analysis charts, and final insights.

9. CHALLENGES FACED

Handling missing values, understanding dataset structure, and creating meaningful visualizations.

10. LEARNING OUTCOMES

Improved Python programming, data analysis, visualization skills, and practical experience with real-world datasets.

11. CONCLUSION

The project successfully analyzed student performance data and generated insights that can support educational improvement and decision-making.